



12493 - Through a Circumstellar Disk, Darkly, with JWST

Cycle: 4, Proposal Category: DD

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRSPEC spectroscopy				
	1		NIRSpec IFU Spectroscopy	(1) ASASSN-25cd
MIRI Spectroscopy				
	2		MIRI Medium Resolution Spectroscopy	(1) ASASSN-25cd

ABSTRACT

We propose time-critical NIRSpec and MIRI observations of an enigmatic K giant star that has achromatically dimmed by over 3 magnitude in the optical since March 2025. The shape of the light curve and absence of any infrared excess rule out a simple eclipsing binary or a rho Corona Borealis-type variable and suggests an eclipse of the primary by a cold, optically thick disk around a fainter companion. The system is too old (Gyrs) for the disk to be primordial and the disk is too thick for a standard collisional debris disk, hinting at a novel pathway for disk formation in evolved systems. A SPHEREx 1-5 micron spectrum obtained 6 months into the event contains a the 3-micron absorption band of water ice and possible features of CO₂ and CO, indicating a disk composed of large, icy grains. We propose NIRSPEC and MIRI spectroscopy to exploit this unparalleled opportunity to investigate a novel class of circumstellar disk. The greater sensitivity and spectral resolution will allow us to identify

JWST Proposal 12493 (Created: Tuesday, February 24, 2026, 5:00:17PM Eastern Standard Time) - Overview

lines of gases and analyze ice band profiles in order to (1) compare the volatile/ice inventory to that of protoplanetary disks; (2) constrain the conditions of formation of the grains; and (3) evaluate possible scenarios for the origin of the disk, i.e. mass transfer from the primary star vs. disintegration of small bodies driven by the primary's luminosity evolution. JWST is the only observatory with the required spectral reach and sensitivity, and DD time is critical to obtaining these observations before the end of the eclipse, which cannot be accurately predicted but based on other, similar events could occur before the start of the next GO opportunity in July 2027.

OBSERVING DESCRIPTION

JWST DD observations of the anomalous variable star ASASSN-25cd will consist of high-resolution spectroscopy with NIRSpec in IFU mode covering 1.8-5 micron using two disperser/filter settings (G235H/F170LP, and G395H/F290LP) and MIRI in MRS mode to cover 5-28 microns using all three grating settings and four channels.

Proposal 12493 - Targets - Through a Circumstellar Disk, Darkly, with JWST

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	ASASSN-25cd	RA: 18 31 52.7800 (277.9699167d)	Proper Motion RA: -0.88 mas/yr	
			Dec: -28 48 27.14 (-28.80754d)	Proper Motion Dec: 2.86 mas/yr	
			Equinox: J2000	Epoch of Position: 2000	
	Comments: Category=Star Description=[Circumstellar disks, K giants]				

Proposal 12493 - Observation 1 - Through a Circumstellar Disk, Darkly, with JWST

Observation	Proposal 12493, Observation 1											Tue Feb 24 22:00:17 GMT 2026
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 1:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	ASASSN-25cd	RA: 18 31 52.7800 (277.9699167d) Dec: -28 48 27.14 (-28.80754d) Equinox: J2000				Proper Motion RA: -0.88 mas/yr Proper Motion Dec: 2.86 mas/yr Epoch of Position: 2000					
Comments: Category=Star Description=[Circumstellar disks, K giants]												
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-NOD										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	19	2	false	true	NONE	4	8	2334.222	278195
	2	G235H/F170LP	NRSIRS2RAPID	19	2	false	true	NONE	4	8	2334.222	278195
Special Requirements	Between Dates 23-MAR-2026:00:00:00 and 12-MAY-2026:00:00:00											

Proposal 12493 - Observation 2 - Through a Circumstellar Disk, Darkly, with JWST

Observation	Proposal 12493, Observation 2												Tue Feb 24 22:00:17 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(Visit 2:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	ASASSN-25cd	RA: 18 31 52.7800 (277.9699167d)				Proper Motion RA: -0.88 mas/yr						
			Dec: -28 48 27.14 (-28.80754d)				Proper Motion Dec: 2.86 mas/yr						
			Equinox: J2000				Epoch of Position: 2000						
	Comments: Category=Star Description=[Circumstellar disks, K giants]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction	
		All MRS				YES				FULL		Allow Auto Reorder	
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				POSITIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	SLOWR1	25	1	1	Dither 1	4	4	2388.992	278195
	1	SHORT(A)	MRSLONG		SLOWR1	25	1	1	Dither 1	4	4	2388.992	278195
	1	SHORT(A)	MRSSHORT		SLOWR1	25	1	1	Dither 1	4	4	2388.992	278195
	2		IMAGER	F770W	SLOWR1	25	1	1	Dither 1	4	4	2388.992	278195
	2	MEDIUM(B)	MRSLONG		SLOWR1	25	1	1	Dither 1	4	4	2388.992	278195
	2	MEDIUM(B)	MRSSHORT		SLOWR1	25	1	1	Dither 1	4	4	2388.992	278195
	3		IMAGER	F770W	SLOWR1	25	1	1	Dither 1	4	4	2388.992	278195
	3	LONG(C)	MRSLONG		SLOWR1	25	1	1	Dither 1	4	4	2388.992	278195
	3	LONG(C)	MRSSHORT		SLOWR1	25	1	1	Dither 1	4	4	2388.992	278195

Proposal 12493 - Observation 2 - Through a Circumstellar Disk, Darkly, with JWST

Special Requirements	Between Dates 23-MAR-2026:00:00:00 and 12-MAY-2026:00:00:00
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